COMPETENCY STANDARD

FOR

WEAVING AND FINISHING
(JUTE SECTOR)

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-WVF-01-G	Apply occupational health and safety (OHS) practices at workplace	1. Identify, control and report OHS hazards 2. Conduct work safely 3. Follow emergency response procedures 4. Maintain and improve health and safety in the workplace	10
SICIP- JUTE-WVF-02-G	Work In a Team Environment	1. Identify team goals and work processes 2. Communicate and co-operate with team members 3. Work as a team member 4. Solve problems as a team member	10
		Total Hour	**20**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-WVF-01-S	Apply Quality Procedures in Jute Manufacturing	1. Identify quality procedures 2. Follow quality procedures 3. Maintain standard procedures	10
SICIP- JUTE-WVF-02-S	Apply green practices in Jute sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**20**

Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-JUTE-WVF-01-O	Demonstrate Knowledge on Jute Weaving and Finishing	1. Describe role and responsibilities of a weaving operator 2. Interpret principles and types of jute weaving 3. Identify types of jute fabric, yarn and their end-uses 4. Illustrate the operational flow of weaving and finishing processes	25 Hrs.
SICIP-JUTE-WVF-02-O	Prepare Loom for Weaving Operation	1. Inspect and prepare warp beam and yarn for loom 2. Set up heald wire, reed and drop wires 3. Draw-in and tie-in warp threads 4. Set weaving parameters 5. Conduct trial run and adjust loom setting	35 Hrs.
SICIP-JUTE-WVF-03-O	Operate Weaving Loom and Produce Jute Fabric	1. Start up and operate the weaving loom 2. Monitor weft insertion and shed formation 3. Monitor fabric formation and detect faults 4. Perform in-process quality checks 5. Stop, doff and remove woven fabric roll	45 Hrs.
SICIP-JUTE-WVF-04-O	Monitor and Control Weaving Quality and Loom Performance	1. Identify and rectify common weaving faults 2. Monitor loom efficiency and production output 3. Record production quality wise 4. Coordinate with maintenance for breakdown and quality issues	30 Hrs.
SICIP-JUTE-WVF-05-O	Perform Fabric and Product Finishing Operations	1. Prepare fabric for finishing 2. Perform calendering and pressing operations 3. Perform cutting and sewing operations 4. Perform lapping and baling of finished goods 5. Survey of finish goods	45 Hrs.
		Total	**180**

COMPETENCY STANDARD: WEAVING AND FINISHING

Generic Competencies:

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES AT WORKPLACE**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-JUTE-WVF-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices at workplace. It specifically includes the tasks of identifying, controlling and reporting OHS hazards, conducting work safely, following emergency response procedures and maintaining and improving health and safety in the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify, control and report OHS hazards	1.1 Work area is routinely checked for Occupational Health and Safety (OHS) hazards prior to commencing and during work. 1.2 **<u>Hazards</u>** are identified and corrective action is taken within the level of responsibility. 1.3 OSH hazards and incidents are reported to appropriate personnel according to workplace procedures. 1.4 **<u>Safety signs and symbols</u>** are identified and followed. 1.5 **<u>Preventive Control measures</u>** are identified in accordance with OSH work standards.
2. Conduct work safely	2.1 OHS practices are applied in the workplace. 2.2 **<u>Personal Protective Equipment (PPE)</u>** are used.
3. Follow emergency response procedures	3.1 Emergency situations are identified and reported according to workplace requirements. 3.2 Emergency procedures are followed as appropriate to the nature of the emergency and according to workplace procedures. 3.3 **<u>Workplace procedures</u>** for dealing with accidents, fires and emergencies are followed whenever necessary within scope of responsibilities.
4. Maintain and improve health and safety in the workplace	4.1 Risks are identified and appropriate control measures are implemented in the workplace. 4.2 Recommendations arising from risk assessments are

	implemented within level of responsibility. 4.3 Safety records are maintained according to **<u>company policies</u>.**

Range of Variables

Variable	**Range** (May include but not limited to):
1. Hazards	1.1. OHS incidents include near misses, injuries, illnesses and property damage, noise, handling hazardous substances. 1.2. Biological hazards- bacteria, viruses, plants, parasites, mites, molds, fungi, insects 1.3. Chemical hazards – dusts, fibers, mists, fumes, smoke, gasses, vapors 1.4. Ergonomics Hazards 1.5. Physiological factors – monotony, personal relationship, work out cycle 1.6. Safety hazards (unsafe workplace condition) – confined space, excavations, falling objects, gas leaks, electrical, poor storage of materials and waste, spillage, waste and debris 1.7. Unsafe workers' act (Smoking in off-limited areas, Substance and alcohol abuse at work) 1.8. Working with and near moving equipment / load shifting equipment. 1.9. Broken or damaged equipment or materials.
2. Safety signs and symbols	2.1 Direction signs (exit, emergency exit, etc.) 2.2 First aid signs 2.3 Danger Tags 2.4 Hazard signs 2.5 Safety tags 2.6 Warning signs
3. Personal Protective Equipment (PPE)	3.1 Apron 3.2 Safety Helmet 3.3 Goggles 3.4 Ear muffs 3.5 Ear plugs 3.6 Gloves 3.7 Clothing 3.8 Safety Boots 3.9 Face Protection (Musk & Shield) 3.10 Scarf 3.11 Hair Net/cap/bonnet

	3.12 Hard hat
4. Preventive Control Measures	4.1 Eliminate the hazard (i.e., get rid of the dangerous machine 4.2 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 4.3 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 4.4 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OHS-related topics, issue warning signage, rotation/shifting work schedule) 4.5 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 4.6 Use personal protective equipment 4.7 Safety, health and work environment evaluation 4.8 Periodic and/or special medical examinations of workers
5. Workplace Procedures	5.1 Fire fighting 5.2 Medical and first aid 5.3 Evacuation 5.4 Material Safety Data Sheets (MSDSs) and manufacturers' advice
6. Company policies	6.1 Job related Standard Operating Procedures (SOPs). 6.2 Occupational Health and Safety (OHS) specific procedures. 6.3 Examples of OHS procedures include – consultation and participation, emergency response to specific hazards, incident investigation, risk assessment, reporting arrangement and issue resolution procedures.

Curricular Content Guide

1. Underpinning Knowledge	1.1. OHS Workplace Policies and Procedures. 1.2. Work safety procedures. 1.3. Fire and emergency procedures. 1.4. Types of Hazards (Biological, Chemical and Physical) and their effects. 1.5. PPE types and uses. 1.6. Personal hygiene practices. 1.7. OHS awareness. 1.8. Steps of hazard identification. 1.9. Principles of hazards control. 1.10. Employer's role.

	1.11. Supervisor's responsibilities.
2. Underpinning Skills	2.1. Identifying OHS policies and procedures. 2.2. Following personal work safety practices. 2.3. Reporting hazards and risks. 2.4. Responding to emergency procedures. 2.5. Maintaining physical well-being in the workplace. 2.6. Identify tools and equipment related to OHS. 2.7. Improving OHS performance.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Manuals 4.4 Workplace documents, signs and symbols 4.5 Relevant specifications or work instructions. 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 used Personal Protective Equipment (PPE). 1.2 identified hazards. 1.3 took corrective action of different hazards. 1.4 took corrective action for emergency procedure. 1.5 reported emergency situation to the Supervisor / Manager. 1.6 satisfied requirements mentioned in the performance criteria and range of variables.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated

	work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 Hrs**.**	**Unit Code:** SICIP-JUTE-WVF-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work in a team environment. It specifically includes the tasks of Identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes.	1.1. Team goals and collaborative decision-making processes are identified. 1.2. Role and common goals of the team are defined from available **<u>sources of information.</u>** 1.3. Team structure, responsibilities and reporting relations are identified from team discussions and other external sources.
2. Communicate and co-operate with team members.	2.1 Communication and negotiation skills are applied and maintained in all relevant situations. 2.2 Constructive contributions are made to **<u>workplace discussions</u>** on such issues as production, quality and safety. 2.3 Goals/ objectives and action plans undertaken in the workplace are communicated promptly. 2.4 Information regarding problems and issues are organized coherently to ensure clear and effective communication. 2.5 Dialogue is initiated with appropriate personnel. 2.6 Communication problems and issues are raised 2.7 Barriers to communication are identified and resolved
3. Work as a team member	3.1 Effective forms of communication are used to interact with **<u>team members</u>** in discussing team activities and objectives. 3.2 Mutual respect, empathy and active collaboration are

	demonstrated 3.3 Communication channels are followed as per **<u>workplace context.</u>**
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified 4.2 Problems are investigated and analyzed 4.3 Potential solutions of problem are identified 4.4 Recommendations about possible solutions are developed, documented, ranked and presented to team members for decision.

Range of Variables

Variable	**Range** (May include but not limited to):
1. Sources of information	1.1. Organizational structures 1.2. Operations Manuals 1.3. Job description 1.4. Standard operating procedures
2. Workplace discussions	2.1 Coordination meetings 2.2 Toolbox discussion 2.3 Peer-to-peer discussion
3. Team members	3.1 Coach / members 3.2 Supervisor / manager 3.3 Peers / colleagues 3.4 Other members /Employee representative of the organization.
4. Workplace context	4.1 National laws and statutes 4.2 Standard operating procedure 4.3 Workplace rules and regulations

Curricular Content Guide

1. Underpinning Knowledge	1.1. Sources of information define 1.2. Team structure, role, and responsibility. 1.3. Individual member's roles and responsibilities. 1.4. Effective verbal communication methods 1.5. Communication flow and reporting structures. 1.6. Interpersonal communication skills. 1.7. Organization requirements for written and electronic communication methods 1.8. Communication problems and issues 1.9. Barriers in communication 1.10. Team planning. 1.11. Team meeting procedures. 1.12. Workplace etiquette

	1.13. Industry maintenance, service and helpdesk practices, processes and procedures 1.14. Industry standard diagnostic tools 1.15. Malfunctions and resolutions
2. Underpinning Skills	2.1 Organizing sources of information 2.2 Identifying the role and responsibility of the team. 2.3 Identifying roles and responsibilities of individual members. 2.4 Identifying effective verbal communication methods 2.5 Identifying communication flow and reporting structure. 2.6 Identifying interpersonal communication skills 2.7 Complying with organization requirements for the use of written and electronic communication methods 2.8 Negotiation and communication skills 2.9 Participating in team discussion. 2.10 Working as a team member. 2.11 Participating in a variety of workplace discussions 2.12 Effective clarifying and probing skills 2.13 Identifying issues 2.14 Identifying current industry standard diagnostic tools 2.15 Describing common malfunctions and resolutions. 2.16 Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Relevant specifications or work instructions. 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes

	1.2 identified own role and responsibilities within team 1.1 communicated and cooperated with team members 1.2 practice problem solving within the team
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Sector Specific Competencies (20 Hrs.)

Unit of Competency: **APPLY QUALITY PROCEDURES IN JUTE MANUFACTURING**	**Nominal Duration:** 10 Hrs**.**	**Unit Code:** SICIP-JUTE-WVF-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply quality procedures in jute manufacturing. It specifically includes the tasks of identifying quality procedures, following quality procedures and maintaining standard procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify quality procedures	1.1. **Manuals** and guidelines for jute production processes are collected as per product specifications. 1.2. The importance of manuals in ensuring quality standards is recognized. 1.3. Instructions and procedures for quality control are identified. 1.4. Required information is collected from manuals. 1.5. Performance measurement systems are identified.
2. Follow quality procedures	2.1 Instructions and procedures are followed strictly and duties are performed in accordance with the **quality improvement system**. 2.2 Concept of supplying jute products to meet **customer quality requirements** is understood and applied. 2.3 Conformance to specifications is ensured.

	2.4 Defects and deviations are detected and reported to the relevant authority according to standard operating procedures.
3. Maintain standard procedures	3.1 Performance is assessed at regular intervals. 3.2 Specifications and standard operating procedures are established. 3.3 Quality of jute product is checked and verified. 3.4 Quality control and quality assurance system procedures for each job are followed. 3.5 Conformance to specification is ensured in all situations.

Range of Variables

Variable	Range (May include but not limited to):
1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual 1.3 Jute product quality manual 1.4 Maintenance procedure manual 1.5 Signs and symbols, instruction manuals
2. Quality improvement system	2.1 Quality inspection 2.2 Testing 2.3 Quality control 2.4 Quality assurance 2.5 Total Quality Management (TQM)
3. Customer quality requirements	3.1 Performance 3.2 Features 3.3 Reliability 3.4 Conformance 3.5 Aesthetics 3.6 Durability

Curricular Content Guide

1. Underpinning Knowledge	1.1 Themes on various types of RMG manuals 1.2 Schedules, dimensions and specifications contained in sketch 1.3 Units of conversion 1.4 Rules of sketch, drawings and specifications 1.5 Sketch and specifications 1.6 Signs and symbols 1.7 Terms and abbreviations used
2. Underpinning Skills	2.1 Recognising importance of manual 2.2 Selecting appropriate manuals as per sample 2.3 Collecting information from manual as per sample

	2.4 Identifying signs and symbols 2.5 Identifying drawings and specifications 2.6 Interpreting schedules, dimensions, drawings and specifications
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals 4.3 Workplace documents, signs and symbols 4.4 Sketch and specifications 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected information from manual as per sample 1.2 Identified relevant drawings and specifications 1.3 identified signs and symbols 1.4 identified terms and abbreviations 1.5 identified sketches and specifications as per sample
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN JUTE SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-JUTE-WVF-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in Jute sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **Principles of green practices in Jute** manufacturing are explained. 1.2 Key terms and symbols in Jute production processes are identified. 1.3 **Sources of environmental impacts** during Jute manufacturing activities are identified. 1.4 Jute manufacturing activities use are explained. 1.5 **Ways of mitigating environmental impacts** in Jute manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in Jute manufacturing are documented. 2.2 **Recyclable and non-recyclable items** in Jute production processes are identified. 2.3 Procedures to reduce resource consumption in Jute manufacturing are implemented. 2.4 Sustainable alternatives in Jute manufacturing are explored and applied.
3. Implement waste management practices	3.1 **Different types of waste in Jute manufacturing** are identified. 3.2 Hazardous waste in jute production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range (May include but not limited to):
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in Jute manufacturing processes. 1.2 6R for waste management in Jute production: 1.2.1 Refuse unnecessary materials and processes.

	1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in Jute manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in Jute production with low environmental impact. 1.4 Recycling of materials used in the Jute manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the Jute sector.
2. Sources of environmental impacts	2.1 Air pollution in Jute manufacturing 2.2 Water pollution in Jute manufacturing 2.3 Soil degradation in Jute manufacturing 2.4 Noise pollution in Jute manufacturing 2.5 Waste generation in Jute manufacturing 2.6 Resource depletion in Jute manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in Jute production processes. 3.2 Adopting renewable energy sources for Jute manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 **Recyclable Items in RMG Manufacturing** 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 **Non-Recyclable Items in Jute Manufacturing** 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.

5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Jute manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in Jute manufacturing 1.6 Hazardous waste in Jute production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in Jute manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in Jute manufacturing 2.6 Disposing hazardous waste in Jute production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure

	4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation Specific Competencies (180 Hrs.)

Unit of Competency:	Nominal Duration:	Unit Code:
DEMONSTRATE KNOWLEDGE ON JUTE WEAVING AND FINISHING	25 Hrs.	SICIP-JUTE-WVF-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to demonstrate foundational knowledge of jute weaving and finishing. It specifically includes the tasks of describing the role and responsibilities of a weaving and finishing operator, interpreting the principles and types of jute weaving, identifying types of jute fabric and yarn and their end-uses, and illustrating the operational flow of weaving and finishing processes.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Describe role and responsibilities of a weaving operator	1.1 Job role and scope of work of a weaving operator are described in accordance with industry practice. 1.2 Duties and responsibilities within the weaving section are identified as per organizational structure. 1.3 Reporting relationships and lines of communication in the weaving shed are explained. 1.4 Personal conduct, discipline and work ethics required in the section are described. 1.5 Career progression pathways within the jute weaving trade are identified.
2. Interpret principles and types of jute weaving	2.1 Basic principles of weaving (interlacement between warp and weft) are explained. 2.2 **Types of weaves** are identified. 2.3 Classification of **jute looms** is described. 2.4 Difference between hessian and sacking construction is explained. 2.5 Basic **weaving terminologies** are defined.
3. Identify types of jute fabric, yarn and their end-uses	3.1 **Types of jute yarn** are identified. 3.2 **Types of jute fabric** are identified. 3.3 End-uses of each **fabric and bags type** are described. 3.4 Standard **fabric specifications** are interpreted from product specification sheets.
4. Illustrate the operational	4.1 Sequential flow of pre-weaving operations to weaving is

Elements of Competency	Performance Criteria
flow of weaving and finishing processes	illustrated. 4.2 Weaving process flow from loom setting to fabric doffing is illustrated. 4.3 **<u>Finishing process flow</u>** is illustrated. 4.4 Interrelationship between weaving and finishing sections is explained using a flow chart.

Range of Variables

Variable	Range (may include but not limited to)
1. Types of weaves	1.1 Plain 1.2 Twill
2. Jute looms	2.1 Hessian loom 2.2 Sacking loom 2.3 Power loom 2.4 Shuttle less loom
3. Weaving terminologies	3.1 Warps/Ends 3.2 Wefts/Picks 3.3 Reed 3.4 Heald 3.5 Shed 3.6 Selvedge
4. Types of jute yarn	4.1 Warp yarn 4.2 Weft yarn
5. Types of jute fabric	5.1 Hessian cloth 5.2 Sacking cloth 5.3 Carpet Backing Cloth (CBC) 5.4 Diversified Jute Products (DJP) fabric
6. Fabric and bags type	6.1 Packaging, Agriculture 6.2 Geo-Jute Textiles 6.3 Diversified Products
7. Fabric specifications	7.1 Width 7.2 Weight Per Unit Area 7.3 Pick Density
8. Finishing process flow	8.1 Damping

Variable	Range (may include but not limited to)
	8.2 Calendaring 8.3 Cutting 8.4 Sewing 8.5 Baling

Curricular Content Guide

1. Underpinning Knowledge	1.1 Organizational structure and job roles within the weaving and finishing section 1.2 Basic principles of weaving (interlacement of warp and weft threads) 1.3 Classification of jute looms and their operating principles 1.4 Types of jute fabric and their industrial/commercial end-uses 1.5 Standard weaving terminologies 1.6 Overview of pre-weaving preparatory processes (batching, spinning, winding, beaming) 1.7 Overview of finishing processes (damping, calendaring, cutting, sewing, baling) 1.8 Workplace communication and reporting structure 1.9 Basic occupational health and safety requirements in the weaving/finishing shed 1.10 Career pathways and progression opportunities in the jute weaving trade
2. Underpinning Skills	2.1 Identifying roles, responsibilities and reporting lines within the section 2.2 Interpreting basic weaving principles and terminology 2.3 Distinguishing between types of jute looms and fabrics 2.4 Reading and interpreting product/fabric specification sheets 2.5 Illustrating process flow charts for weaving and finishing operations 2.6 Communicating effectively with supervisors and team members 2.7 Applying basic safety practices in the work area 2.8 Using reference manuals and organizational documents to gather information 2.9 Recognizing standard fabric types by visual and physical characteristics
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual weaving/finishing section) 4.2 Sample fabric and yarn specimens 4.3 Reference manuals and product specification sheets 4.4 Organizational charts and SOPs 4.5 Audio-visual aids/process flow charts

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described roles and responsibilities of a weaving/finishing operator 1.2 explained principles and types of jute weaving 1.3 identified types of jute fabric, yarn and their end-uses 1.4 illustrated the operational flow of weaving and finishing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PREPARE LOOM FOR WEAVING OPERATION**	**Nominal Duration:** 35 Hrs**.**	**Unit Code:** SICIP-JUTE-WVF-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to prepare a loom for weaving operation. It specifically includes the tasks of inspecting and preparing the warp beam and yarn, setting up healds, reed and drop wires, drawing-in and tying-in warp threads, setting weaving parameters and conducting a trial run to adjust loom settings.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect and prepare warp beam and yarn for loom	1.1 Warp beam is checked for correct yarn count, length and kill marks as per work order. 1.2 Yarn breakages and **<u>defects</u>** on the warp beam are identified and reported. 1.3 Warp beam is mounted onto the loom back rail using appropriate lifting technique/equipment. 1.4 Beam tension is set and verified in accordance with the machine manual. 1.5 Required weft yarn (spools/cops) is prepared
2. Set up heald wire, reed and drop wires	2.1 Healds are selected and mounted according to the design and drafting plan. 2.2 Reed of correct count/dent is selected and fitted as per fabric specification. 2.3 Droppers are threaded and positioned to detect warp breakages. 2.4 Heald frames are inspected for damage and replaced/repaired as required.
3. Draw-in and tie-in warp threads	3.1 Warp threads are drawn through drop wires, healds and reed following the drafting plan. 3.2 Drawn-in ends are checked against the design/pattern card for accuracy. 3.3 Tie-in of new warp to old warp is performed, where a beam change is required, following standard knotting technique. 3.4 Missing or crossed ends are identified and corrected before loom start-up. 3.5 Warp threads are evenly tensioned across the full width of the beam.
4. Set weaving parameters	4.1 **<u>weaving parameters</u>** are set according to fabric specification and work order. 4.2 Let-off and take-up motions are adjusted to maintain uniform fabric width and pick density. 4.3 Weft insertion mechanism (shuttle/gripper) is set and

Elements of Competency	Performance Criteria
	checked for correct operation. 4.4 Selvedge device is set to produce a firm and even selvedge. 4.5 Loom settings are recorded on the production/bin card.
5. Conduct trial run and adjust loom setting	5.1 Loom is run at low speed to check for correct shedding, picking and beat-up. 5.2 Trial fabric is inspected for weave pattern, width and defects. 5.3 Adjustments are made to tension, timing or settings based on trial run results. 5.4 Loom is confirmed ready for production and handed over as per workplace procedure.

Range of Variables

Variable	Range (May include but not limited to):
1. Defects	1.1 Loose beam 1.2 Tight beam 1.3 Numbers of excess or short ends 1.4 Beam without kill marks
2. Weaving parameters	2.1 Weft Count, 2.2 Loom Speed, 2.3 Weft Fork Motion 2.4 Shed Timing

Curricular Content Guide

1. Underpinning Knowledge	1.1 Warp beam preparation and mounting procedures 1.2 Drafting and denting plans for healds and reed 1.3 Heald and reed selection criteria based on fabric design 1.4 Drop wire function and warp breakage detection 1.5 Drawing-in and tying-in techniques and standard knots 1.6 Loom setting parameters (pick count, speed, shed timing, tension) 1.7 Let-off and take-up motion functions 1.8 Selvedge formation requirements 1.9 Trial run procedures and fault-checking methods 1.10 Relevant machine manuals and standard operating procedures
2. Underpinning Skills	2.1 Inspecting and mounting warp beams safely and

	correctly 2.2 Selecting and fitting healds, reed and drop wires per design 2.3 Drawing-in and tying-in warp threads accurately 2.4 Setting loom parameters according to specification 2.5 Adjusting let-off and take-up motions 2.6 Conducting trial runs and interpreting results 2.7 Identifying and correcting drawing-in/threading errors 2.8 Using hand tools and lifting equipment safely 2.9 Recording machine settings on production/setting cards
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Weaving loom (shuttle/power loom) 4.2 Warp beam, healds, reed, drop wires 4.3 Drafting/denting plan 4.4 Hand tools and knotting tools 4.5 PPE appropriate to the activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 inspected and mounted warp beam correctly 1.2 set up healds, reed and drop wires as per design 1.3 drew-in and tied-in warp threads accurately 1.4 set weaving parameters according to specification 1.5 conducted trial run and adjusted loom setting
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified

	assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE WEAVING LOOM AND PRODUCE JUTE FABRIC**	**Nominal Duration:** 45 Hrs.	**Unit Code:** SICIP-JUTE-WVF-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate a weaving loom and produce jute fabric. It specifically includes the tasks of starting up and operating the loom, monitoring weft insertion and shed formation, monitoring fabric formation and detecting faults, performing in-process quality checks, and stopping, doffing and removing the woven fabric roll.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Start up and operate the weaving loom	1.1 Pre-start safety checks are carried out following OHS procedures. 1.2 Loom is started following standard operating procedure. 1.3 Running speed is monitored and maintained within the specified range. 1.4 Loom operation is continuously observed for abnormal sound, vibration or stoppage. 1.5 Machine stoppages are attended to promptly to minimize downtime.
2. Monitor weft insertion and shed formation	2.1 Weft supply (cop/ spools) is replenished before yarn exhaustion. 2.2 Weft breakages are detected and rectified following standard procedure. 2.3 Shed formation is observed for correct timing and clearance. 2.4 Selvedge formation is monitored for uniformity and firmness.
3. Monitor fabric formation and detect faults	3.1 Woven fabric is visually inspected at regular intervals during production. 3.2 **<u>Common weaving faults</u>** are identified. 3.3 Cause of detected faults is determined and corrective action is taken within scope of responsibility. 3.4 Faults beyond the operator's scope are reported to the supervisor/technician according to workplace procedure.

Elements of Competency	Performance Criteria
	3.5 Fault occurrences are recorded in the production log.
4. Perform in-process quality checks	4.1 Fabric **quality** are checked against specification at set intervals. 4.2 Sample cuttings are taken and checked using appropriate **measuring instruments**. 4.3 In-process quality records are maintained as per standard operating procedure. 4.4 Non-conforming fabric is segregated and reported according to workplace quality procedure.
5. Stop, doff and remove woven fabric roll	5.1 Loom is stopped safely following shut-down procedure when the fabric roll reaches specified length. 5.2 Woven fabric roll is doffed from the take-up roller using correct technique. 5.3 Doffed roll is labeled with production details (loom number, date, length, quality code). 5.4 Fabric roll is transported and handed over to the finishing section as per workplace procedure. 5.5 Loom is prepared and restarted for the next production cycle.

Range of Variables

Variable	Range (may include but not limited to)
1. Common weaving faults	1.1 Reed mark 1.2 Missing end 1.3 Weft bar 1.4 Float 1.5 Thin place 1.6 Thick place 1.7 Oil stain 1.8 Weft crack 1.9 Gaw 1.10 Bias
2. Quality	2.1 Width 2.2 Weft Count 2.3 Weight
3. Measuring	3.1 Magnifying glass

Variable	Range (may include but not limited to)
instruments	3.2 Poter gauge 3.3 Measuring Steel tape 3.4 Weighing balance

Curricular Content Guide

1. Underpinning Knowledge	1.1 Standard operating procedure for loom start-up and shutdown 1.2 Weft insertion mechanisms (shuttle, rapier, pirn/bobbin systems) 1.3 Shed formation and beat-up mechanism principles 1.4 Common weaving fault types, causes and identification 1.5 Fabric inspection methods and measuring instruments 1.6 Selvedge quality requirements 1.7 Production recording and reporting procedures 1.8 OHS requirements for operating weaving machinery 1.9 Basic loom mechanics and moving parts 1.10 Fabric doffing and roll handling procedures
2. Underpinning Skills	2.1 Starting up, operating and shutting down the loom safely 2.2 Monitoring weft insertion and replenishing weft supply 2.3 Detecting and rectifying weft/warp breakages 2.4 Identifying and classifying common weaving faults 2.5 Taking corrective action within scope of responsibility 2.6 Using measuring instruments (pick glass, scale, weighing balance) 2.7 Maintaining in-process production and quality records 2.8 Doffing and handling woven fabric rolls correctly 2.9 Reporting faults and machine issues to supervisor/technician
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Operating weaving loom 4. 2 Weft supply (shuttle/pirn/bobbin)

	4. 3 Measuring instruments (pick glass, scale, weighing balance) 4. 4 Production log/register 4. 5 PPE appropriate to the activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared for repair/replacement work 1.2 dismantled machine components 1.3 repaired or replaced defective components 1.4 reassembled and reinstall components 1.5 tested and commissioned repaired machine
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MONITOR AND CONTROL WEAVING QUALITY AND LOOM PERFORMANCE	Nominal Duration: 30 Hrs.	Unit Code: SICIP-JUTE-WVF-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to monitor and control weaving quality and loom performance. It specifically includes the tasks of identifying and rectifying common weaving faults, monitoring loom efficiency and production output, recording production and quality data, and coordinating with maintenance for breakdown and quality issues.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and rectify common weaving faults	1.1 Causes of common weaving faults are identified 1.2 Faults arising from yarn, loom setting, loom tuning or machine condition are distinguished. 1.3 Corrective action for identified faults is carried out within the operator's level of responsibility.

Elements of Competency	Performance Criteria
	1.4 Preventive measures to minimize recurrence of faults are applied. 1.5 Persistent or major faults are escalated to the supervisor/quality personnel.
2. Monitor loom efficiency and production output	2.1 Loom running time, stoppage time and production output are recorded per shift. 2.2 Loom efficiency is calculated using the standard formula 2.3 **Causes of low efficiency** are analyzed. 2.4 Actions to improve loom efficiency are identified and implemented within scope of responsibility.
3. Record production quality wise	3.1 **Production data** are recorded accurately in the production register/system. 3.2 Quality inspection results are documented as per standard reporting format. 3.3 Records are maintained in accordance with company documentation procedures. 3.4 Production and quality reports are submitted to the supervisor within the specified timeframe.
4. Coordinate with maintenance for breakdown and quality issues	4.1 Machine breakdowns are reported promptly to maintenance personnel following workplace procedure. 4.2 Relevant information (fault symptoms, loom number, time of occurrence) is communicated clearly to the maintenance team. 4.3 Coordination is maintained with maintenance staff during repair and restart of the loom. 4.4 Feedback on recurring mechanical issues is shared with the supervisor for corrective planning.

Range of Variables

Variable	Range (May include but not limited to):
1. Causes of low efficiency	1.1 Stoppages 1.2 Breakages 1.3 Speed loss
2. Production data	2.1 Fabric length 2.2 Weight

	2.3 Quality grade

Curricular Content Guide

1. Underpinning Knowledge	1.1 Classification and causes of common weaving faults 1.2 Loom efficiency calculation methods 1.3 Production data recording systems and formats 1.4 Quality inspection and reporting procedures 1.5 Communication and escalation procedures for machine breakdowns 1.6 Preventive measures for minimizing weaving faults 1.7 Standard production and quality documentation requirements 1.8 Coordination procedures with maintenance personnel
2. Underpinning Skills	2.1 Analyzing weaving faults to determine root cause 2.2 Calculating loom efficiency and production output 2.3 Recording accurate production and quality data 2.4 Identifying and reporting recurring machine issues 2.5 Applying corrective and preventive actions for faults 2.6 Communicating clearly with maintenance and supervisory staff 2.7 Compiling and submitting shift/production reports 2.8 Interpreting production and quality trends
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Production and quality registers 4.2 Efficiency calculation formats/tools 4.3 Communication and reporting formats 4.4 Maintenance request forms

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and rectified common weaving faults 1.2 monitored and calculated loom efficiency

	1.3 recorded production and quality data accurately 1.4 coordinated effectively with maintenance personnel
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM FABRIC AND PRODUCT FINISHING OPERATIONS**	**Nominal Duration:** 45 Hrs**.**	**Unit Code:** SICIP-JUTE-WVF-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform fabric and product finishing operations. It specifically includes the tasks of preparing fabric for finishing, performing calendaring and pressing operations, performing cutting, hemming and sewing operations, and performing lapping, bundling, pressing and baling of finished goods.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Prepare fabric for finishing	1.1 Grey fabric rolls are received and checked against production/quality records. 1.2 Fabric moisture content is conditioned/dampened according to finishing requirement and product specification. 1.3 Fabric is joined to the required length as per work order. 1.4 Fabric is inspected for major defects before proceeding to finishing operations.
2. Perform calendering and inspection operations	2.1 Calendering machine is set with correct temperature, pressure and speed for the fabric type. 2.2 Fabrics are fed and passed through calendering rollers and bowl to achieve the required smoothness and finish. 2.3 Calendered fabric is checked for uniform finish, width

Elements of Competency	Performance Criteria
	and appearance. 2.4 Fabric width and finish quality are recorded and adjustments made where deviations occur.
3. Perform cutting and sewing operations	3.1 Fabric/bag panels are cut to the required dimensions using cutting machine/tools as per specification. 3.2 Cut edges/panels are hemmed using a sewing machine following standard stitch specification. 3.3 **Bags** **sewing** are carried out according to product design and buyer specification. 3.4 Stitched products are checked for stitch quality, seam strength and dimensional accuracy. 3.5 Loose threads are trimmed and cropping is performed to achieve a clean finished appearance.
4. Perform lapping and baling of finished goods	4.1 Finished fabric/products are folded/lapped to the required size as per packing specification. 4.2 Products are bundled and tied according to the standard bundle count and weight. 4.3 Bundles are pressed and baled using a baling press to achieve required density and dimensions as per shipping standard. 4.4 Bales are strapped and marked with product and batch identification information.
1. Survey of finish goods	5.1 Finished goods are inspected according to quality standards. 5.2 Product specifications are verified against work requirements. 5.3 Defects are identified and recorded. 5.4 Product quantity is verified according to production records. 5.5 Finished goods are classified according to quality requirements. 5.6 Inspection findings are documented according to workplace procedures.

Range of Variables

Variable	Range (May include but not limited to):
1. Bags	1.1 Side And Bottom Sewing 1.2 Mouth Hemming
2. Sewing	2.1 Hessian

	2.2 Sacking 2.3 Cotton pack 2.4 Wool pack 2.5 DW flower 2.6 Sugar 2.7 Cement

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fabric conditioning and damping requirements for finishing 1.2 Calendaring machine operation and settings 1.3 Cutting and sewing techniques for jute bags/products 1.4 Stitch types and seam specifications used in jute products 1.5 Lapping, bundling and baling procedures and standards 1.6 Baling press operation and safety requirements 1.7 Product specifications for hessian, sacking and diversified jute products 1.8 Packing and bundling weight/dimension standards 1.9 Quality requirements for finished product appearance 1.10 OHS requirements for finishing machinery operation
2. Underpinning Skills	2.1 Conditioning and batching grey fabric for finishing 2.2 Operating calendaring machinery to achieve required finish 2.3 Cutting fabric/panels to specification 2.4 Operating sewing machines to produce quality seams and hems 2.5 Cropping and trimming finished products 2.6 Lapping, bundling and tying finished goods 2.7 Operating baling press to produce compliant bales 2.8 Labeling and marking bales/bundles accurately 2.9 Applying safety procedures when operating finishing equipment
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4. 1 Damping/conditioning and calendaring machines 4. 2 Cutting and sewing machines 4. 3 Baling press and bundling/strapping tools 4. 4 Packing materials and labels 4. 5 PPE appropriate to the activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. 1 prepared fabric for finishing correctly 1. 2 performed calendaring and pressing operations to specification 1. 3 performed cutting, hemming and sewing to standard 1. 4 performed lapping, bundling and baling correctly
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Weaving and Finishing**
Number of Trainees:	30

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)

  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and Tools	Required Facilities
1.	Desktop Computers	15 Nos.
2.	Instructor Computer	1 Nos
3.	Multimedia Projector	1 Nos
4.	Whiteboard and markers	1 Nos
5.	Flip chart stands with paper	2 Nos
6.	Printer	1 Nos
7.	Flatbed Scanner	1 Nos
8.	Dust shaker	1 Nos
9.	Emulsion plant	1 Nos
10.	Softener	1 Nos
11.	Spaeder	1 Nos
12.	Jute Fiber Sample Collection	1 Complete Set
13.	Jute Yarn Sample Collection	1 Complete Set
14.	Jute Fabric Sample Collection	1 Complete Set
15.	Diversified Jute Product Sample Set	1 Complete Set
16.	Fabric and Yarn Testing Kit	5 Sets
17.	GSM Cutter and GSM Balance	2 Sets
18.	Digital Weighing Scale	2 Nos.
19.	Quality Inspection Tools	5 Sets
20.	Measuring Tapes	25 Nos.
21.	Set squires (Large)	30 Nos.
22.	L-squire	30 Nos.
23.	Scale (12’,18’, 24’, 40’)	25 Nos.
24.	Cutter Board (Large size)	2 Nos.
25.	Pattern Board	12 Pics

26.	Scissor	5 Nos.
27.	Design instruments Box (set of 20)	1 Nos.
28.	Anty cutter	7 Nos.
29.	Cutting Table with 6 drawers (3’ Hight) (4’x8’)	1 Nos.
30.	Digital scale	1 Nos
31.	First aid kit	1 Nos
32.	Fire extinguisher (for safety demonstration)	1 Nos

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Weaving and Finishing (Jute sector)** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 30 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 30 trainees.